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PI Name:	Turker, Mitchell Ph.D.		
Project Title:	The Relation Between Mutagenesis and Genomic Instability After Particle Exposure In Vivo		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:			
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No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
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Flight Program:			
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COI Name (Institution):	Grygoryev, Dmytro (Oregon Health & Science University) Thayer, Mathew (Oregon Health & Science University) Kronenberg, Amy (Lawrence Berkeley National Laboratory)		
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Task Description:	Astronauts in the space environment are at risk for a variety of health-related concerns from exposure to accelerated ions (fast moving atoms that have lost their electrons) that can pass through the spacecraft hull and interact with astronauts' cells within the body. These particles are released from the sun in solar particle events (SPE) and exist as part of the Galactic Cosmic Ray (GCR) background radiation, although the types of ions and their energies can differ widely in SPE and GCR. A significant concern for astronaut health is an increased risk of cancer, which is linked to a variety of radiation exposures. The underlying cause of cancer, regardless of its source, is mutations in critical genes and ongoing gains, losses, and rearrangement of chromosomes. Collectively, these ongoing changes in the cell are termed genomic instability. The goal of the proposed work is to determine how frequently different types of accelerated ions found in the GCR cause mutations and genomic instability and to describe what these changes look like. To better extrapolate our results to potential harms to astronauts from space radiation, we will use a mouse model for the proposed work. This will ensure that the cell types of interest found in the spleen and kidney will be exposed to accelerated ions in their normal tissue environment. We will focus on accelerated ions found in space for which relatively little information is currently available. Successful completion of the proposed work will help in the design of long-term travel for humans in space and aid in the protection of space travelers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As described in more detail in the task progress report, we have determined that radiation-induced mutant cells in the mouse kidney epithelium have widespread genomic mutations--far more that we had anticipated before starting this project and more characteristic of cancer cell genomes. This observation suggests that a subset of normal appearing cells in the kidney, and by extension in other epithelial tissues, can persist with extensive radiation-induced damage for long periods of time. We hope to pursue the link between these cells and cancer with future funding.
Task Progress:	Task Progress Final Report -- The Relation Between Mutagenesis and Genomic Instability After Particle Exposure In Vivo (NNX14AC97G) The major goal of the work funded over the last 3 years, and the 4 years of funding before that, has been to identify the types of mutations that are induced by the types of radiation present in the space environment. This work is important for determining the potential risk of cancer for astronauts on prolonged deep space travel, such as a trip to Mars, because they will be exposed to space radiation. By identifying specific types of mutations that are common after space radiation exposure and ultimately determining if these types of mutations are present in tumors induced by space radiation, we will provide a direct link between mutation and cancers that may form as a result of space travel. Assuming this link, our work will apply to essentially every type of cancer caused by space radiation because ionizing radiation clearly induces mutations in mammalian cells and mutations clearly play a key role in initiating cancer formation. A second goal of the funded work has been to determine if mutant cells induced by space radiation exhibit signs of genomic instability, which are ongoing changes to the genome and common in many types of cancer. Scientists who want to study the effects of space radiation on biological system must use the ground based NASA Space Radiation Laboratory (NSRL) located within Brookhaven National Laboratory on Long Island, NY. This accelerator can produce pure charged ions (atoms stripped of electrons) from different atoms -- for example iron (Fe), silicon (Si), oxygen (O), titanium (Ti), hydrogen (protons), and calcium (Ca) – at specific doses, including those found in the space environment. Although these exposures are really simulated space radiation, I will refer to them as space radiation for the purpose of this report. We have examined the mutagenic effect of each of these ions at doses that are relevant for a long space trip or slightly higher. In each case at least one higher dose that would be well outside the range found in the space environment was also tested for comparison. We used an in vivo system, in which mice were exposed to a specific ion at 3 or more doses, because this best mimics what will happen to astronaut tissues. The mice have a genetic marker that allows mutant cells to be selected directly from cells isolated from exposed tissues. Although the mutant cells are then grown in culture to obtain material for molecular and sometimes cytogenetic analyses to determine the types of mutational events induced by exposure, mutation formation occurred in cells in the normal tissue environment within the mouse. We isolated and examined mutant cells from the kidney (epithelial cells) and the spleen (blood T cells). Most work was performed with the kidney epithelial cells. In all cases, we observed increased numbers of mutant cells per tissue after exposure to the ions listed above, and in some cases (Ti and Fe ions) at doses equivalent to approximately one to two particles per cell nucleus. Simply stated, a single particle passing through a cell is the lowest possible exposure, aside from no exposure. For Fe, Ti, O, and Si ions, we observed a linear increase in mutations as a function of dose. The only exception was for Ti ions at the highest dose tested because no increase was observed relative to the next lower dose (7 particles per nucleus relative to 2 particles per nucleus). For protons, no increase was observed at the lower doses tested (0.5 to 3.0 Gy), with a dramatic rise in mutations at 4 and 5 Gy. Average exposures in space for a Mars trip will be approximately 0.2 to 0.4 Gy, but could be higher if a major solar flare occurs, which would release large amounts of protons. The mutant cells we selected were due to mutations that affected mouse chromosome 8. Thus, we focused molecular and cytogenetic studies on this chromosome and identified three types of radiation signature mutations, which are rare in spontaneous mutant cells isolated from unexposed mouse kidneys and relatively common in mutant cells induced by accelerated ion exposure. These are 1) large deletions in which a large piece of chromosome 8 is lost, 2) loss of portions of chromosome 8 that are not linked to loss of the target gene (discontinuous), and 3) translocations in which a portion of chromosome 8 is fused to a piece of another chromosome. Our selection assay only selected for mutational events affecting chromosome 8. To search for evidence of genomic instability, we also examined other chromosomes in the space radiation induced kidney cells for evidence of additional mutations. Although we found mutations affecting other chromosomes in both spontaneous mutants and mutants from the exposed kidneys, these genomic mutations were both quantitatively more frequent and qualitatively distinct in the mutant cells isolated from the irradiated kidneys. A detailed analysis of mutant cells isolated from Fe ion exposed kidneys demonstrated that these cells had very extensive genomic damage, similar to what would be expected in cancer cells. Our goal for future research will be to determine if signature mutations and more extensive genomic damage will be observed in cancers induced by space radiation exposure in the kidney. If so, we will have identified markers that distinguish space radiation induced cancers from those that occur spontaneously, and in doing so will have established an absolute link between mutations and cancers induced by the space environment. This link would then allow our accumulated data to be used to predict cancer risk from space travel at different particle doses and for different tissue types.

	We also examined space radiation mutagenesis in T cells from the spleen. T cells can give rise to leukemias and lymphomas, which are distinct from cancers that occur in epithelial tissues. These cells are more difficult to grow than kidney epithelial cells, so we restricted our analysis to chromosome 8. We identified radiation signature mutations (large deletions and discontinuous mutations), though unlike the kidney cells we did not observe these mutations at the highest doses tested. We believe that this absence occurred because the T cells are more sensitive to radiation exposure and high doses killed most of the mutant cells that were also induced. Interestingly, the kinetics of radiation signature mutation was similar to that reported by another group for space radiation-induced leukemias, suggesting a link between mutations and blood-derived cancers. In sum, we showed that space radiation will induce signature mutations that can be used to qualitatively and quantitatively distinguish spontaneous mutations from those induced by exposure in two distinct cell types, kidney epithelial and spleen T cells. Further we showed that the induced mutant kidney cells had extensive mutations throughout the genome, which was surprising because mutations on chromosome 8 were the only ones required for the mutant cells to be isolated. Because of the known link between mutation and cancer, our work will be useful to estimate cancer risk in astronauts from prolonged space trips.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Norbury JW, Schimmerling W, Slaba TC, Azzam EI, Badavi FF, Baiocco G, Benton E, Bindi V, Blakely EA, Blattnig SR, Boothman DA, Borak TB, Britten RA, Curtis S, Dingfelder M, Durante M, Dynan WS, Eisch AJ, Robin Elgart S, Goodhead DT, Guida PM, Heilbronn LH, Hellweg CE, Huff JL, Kronenberg A, La Tessa C, Lowenstein DI, Miller J, Morita T, Narici L, Nelson GA, Norman RB, Ottolenghi A, Patel ZS, Reitz G, Rusek A, Schreurs AS, Scott-Carnell LA, Semones E, Shay JW, Shurshakov VA, Sihver L, Simonsen LC, Story MD, Turker MS, Uchihori Y, Williams J, Zeitlin CJ. "Galactic cosmic ray simulation at the NASA Space Radiation Laboratory." Life Sci Space Res (Amst). 2016 Feb;8:38-51. https://doi.org/10.1016/j.lssr.2016.02.001 ; PubMed PMID: 26948012 , Feb-2016
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